

I.T.VEDANT

Decode your dreams

Diving Into the World of the Data (DWD)

What is Data?

- Data refers to the raw, unorganized facts and figures that are collected from various sources.
- It can be in the form of numbers, text, images, audio, or other formats, and it forms the building blocks of information and knowledge.

What is Information?

- Information is processed data that has meaning.
- Data becomes information when it is organized or interpreted.

Example:

Data: "100, 200, 150"

Information: "The average sales for the last three months is 150 units."

Data vs Information

Data: Raw and unorganized facts.

Information: Organized data with meaning.

Key Difference: Data is the raw material; information is actionable.

What is Data Analysis?

- Data Analysis is the process of inspecting, cleaning, transforming, and modeling data to discover useful information, inform conclusions, and support decision-making.
- Mainly concerned with answering specific questions from existing data.
- **Tools Used:** Excel, SQL, Tableau, Power BI, Python (for analysis).
- **Typical Tasks:**
  - Summarizing data through statistical measures.
  - Creating reports and dashboards.
  - Providing insights to support business decisions.

What is Data Science?

- Data Science is an interdisciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data.
- Concerned with predicting future trends, identifying patterns, and developing machine learning models.
- **Tools Used:** Python, R, Hadoop, Spark, TensorFlow, etc.
- **Typical Tasks:**
  - Building predictive models using machine learning.
  - Working with unstructured data from various sources.
  - Communicating complex insights through data visualization and storytelling.

Different Steps for Data Analysis

1. **Collect Data:** Identify relevant data sources and gather information.
2. **Explore Data:** Examine the data to understand its structure and identify patterns.
3. **Clean and Transform:** Prepare the data for analysis by addressing missing values, errors, and inconsistencies.
4. **Analyze and Interpret:** Apply statistical techniques to draw insights and conclusions from the data.
5. **Communicate Findings:** Present the analyzed data and its implications to stakeholders.

Data Science Lifecycle

1. **Problem Definition:** Identify the business problem that data science can solve.
2. **Data Collection:** Gather relevant data.
3. **Data Cleaning:** Prepare the data for analysis by handling errors and missing values.
4. **Data Exploration:** Explore and analyze data to uncover patterns.
5. **Model Building:** Apply machine learning algorithms to make predictions.
6. **Model Evaluation:** Test and validate the model's accuracy.
7. **Deployment:** Implement the model in production to provide ongoing insights.

Types of Data Analysis

- **Descriptive Analysis:** Summarizes historical data (e.g., "What happened?").
- **Diagnostic Analysis:** Examines why something happened.
- **Predictive Analysis:** Predicts future outcomes based on historical data.
- **Prescriptive Analysis:** Suggests actions based on predictive analysis (e.g., "What should be done?").

Different Job Roles

- **Data Analyst:** Collects, analyzes, and interprets data to help organizations make informed decisions. Responsible for extracting insights from complex datasets.
- **Data Scientist:** Develops and applies advanced statistical and machine learning techniques to uncover hidden patterns and predict future trends in data.
- **Business Intelligence Analyst:** Transforms data into actionable business intelligence, identifying opportunities and providing strategic recommendations to decision-makers.
- **Data Engineer:** Designs and builds data pipelines, data warehouses, and other infrastructure to ensure data is accessible, secure, and scalable for analysis.

Qualitative vs. Quantitative Data

- **Qualitative Data:** Descriptive, non-numeric information (e.g., colors, opinions, textures).
    - **Example:** Survey responses like "happy," "neutral," "sad."
- **Quantitative Data:** Numeric information that can be measured and counted.
    - **Example:** Height, weight, temperature, or age.

Qualitative Data Types

- **Nominal Data:** Categorical data without a natural order.
    - **Examples:** Gender, color, city names.
- **Ordinal Data:** Categorical data with a meaningful order, but no fixed intervals.
    - **Examples:** Satisfaction ratings (e.g., poor, fair, good), ranks in a competition.

Quantitative Data Types

- **Discrete Data:** Countable data with specific values, usually integers.
    - **Examples:** Number of students, number of cars.
- **Continuous Data:** Data that can take any value within a range and can be measured to any level of precision.
    - **Examples:** Height, temperature, time, and weight.

Structured, Semi-Structured, Unstructured Data

- **Structured Data:** Organized data with a clear format, often stored in tables.
  - **Examples:** Spreadsheets, databases.

- **Unstructured Data:** Lacks a defined structure; harder to process and analyze.
  - **Examples:** Emails, videos, social media posts.

- **Semi-Structured Data:** Partially organized data that doesn't fit a rigid structure but has tags or markers for easier analysis.
  - **Examples:** JSON files, XML data, HTML documents.

What is Statistics?

- Statistics is the science of collecting, analyzing, interpreting, and presenting data.
- Statistical techniques help handle uncertainty, variability, and provide a foundation for model evaluation and hypothesis testing.
- Two Key Branches:
    - **Descriptive Statistics:** Summarizes data features.
    - **Inferential Statistics:** Draws conclusions from data samples.

Descriptive Statistics

- Provides methods to summarize and describe the main features of a dataset.
- **Measure of Central Tendency:** Measures that describe the center point or typical value of a dataset.
    - **Mean (Average):** The sum of all data points divided by the number of points. Useful for understanding the "typical" value.
    - **Median:** The middle value when data points are ordered, which is less sensitive to outliers than the mean.
    - **Mode:** The most frequent value, useful in categorical data.

Problem Statement

A restaurant wants to analyze its daily customer visits to find the average number, the most common number, and the middle number of visits. Given data on daily customer counts over a month, calculate the mean, median, and mode.

Data: [45, 52, 47, 50, 49, 48, 52, 47, 46, 50, 49, 52, 53, 47, 48, 51, 52, 50, 49, 47, 53, 48, 46, 50, 52, 49, 47, 52, 50, 48]

Measures of Dispersion

- Indicates the extent to which data points deviate from the central tendency, providing insights into the variability of the data.

  - **Range:** Measures the spread by subtracting the minimum value from the maximum.

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

  - **Variance (σ^2):** The average of the squared differences from the mean, giving insight into data dispersion.

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$$

  - **Standard Deviation (σ):** The square root of variance, a common measure for the spread of data.

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

  - **Interquartile Range (IQR):** The range between the first quartile (Q1) and the third quartile (Q3) of a dataset. It represents the spread of the middle 50% of data.

$$\text{IQR} = Q3 - Q1$$

Problem Statement

1. A real estate agent wants to analyze house prices in a neighborhood to understand price variability. Calculate the range, variance, standard deviation, and interquartile range (IQR) for the house prices.
   Data: [320000, 340000, 290000, 360000, 310000, 330000, 370000, 300000, 350000, 325000] (in USD)

2. A fitness trainer is analyzing the daily steps taken by a client over two weeks to assess consistency in activity levels. Calculate the range, variance, standard deviation, and interquartile range (IQR) of the daily step counts.
   Data: [7500, 8200, 7800, 7300, 7900, 8400, 8100, 7600, 8000, 8500, 7700, 7900, 8100, 8200]

Measure of Association

- Statistics that describe the degree and direction of relationship between two variables.
  - **Covariance:** Indicates the direction of the relationship between two variables, calculated as the average of the products of their deviations from their means.

$$\mathrm{Cov}(X,Y) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

  - **Correlation:** A statistical measure that indicates the extent to which two variables fluctuate together. Ranges from -1 to +1.
    - **Positive correlation:** As one variable increases, the other tends to increase.
    - **Negative correlation:** As one variable increases, the other tends to decrease.
    - **Zero correlation:** No relationship between the variables.

$$r = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \cdot \sigma_Y}$$

Problem Statement

1. A human resources department wants to understand the relationship between employees' years of experience and their salary levels. Calculate the correlation and covariance to assess if there is a positive relationship between experience and salary.
   Data:
   Years of Experience: [1, 3, 5, 2, 4, 6, 3, 5, 7, 2]
   Salary (in $1000s): [35, 45, 55, 38, 50, 60, 48, 55, 65, 40]

2. A health insurance company is analyzing if there's a relationship between a client's age and their annual health expenditure. Calculate the correlation and covariance to assess the relationship between age and health spending.
   Data:
   Age (in years): [25, 30, 45, 40, 50, 35, 60, 55, 70, 65]
   Health Expenditure (in $1000s): [2.5, 3, 4, 3.5, 5, 3.8, 6, 5.5, 6.8, 6.5]

Probability Basics

- A way to measure uncertainty—the likelihood that an event will happen. It helps in making predictions and understanding model outcomes.
  - Weather Forecasts: Predicting rain with a certain probability.
  - Sports: Calculating chances of winning a game.

- **Random Variable:** A variable whose values are outcomes of a random phenomenon.
  - Example: Rolling a die; the result (1-6) is a random variable.
  - **Types:**
    - **Discrete Random Variable:** Takes on specific values (e.g., rolling a die, which can only land on 1, 2, 3, etc.).
    - **Continuous Random Variable:** Can take any value within a range (e.g., the height of a person).

- **Probability Distributions:** Represent the probabilities of different outcomes; common types include uniform, binomial, and normal distributions.

Probability Basics

- Probability of an Event (E):

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

- **Example:** The probability of rolling a 3 on a fair six-sided die.
  - Favorable Outcome: 1 (rolling a 3).
  - Total Outcomes: 6 (any number from 1 to 6).

$$P(\text{Rolling a 3}) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}} = \frac{1}{6}$$

Problem Statement

1. You are rolling a fair six-sided die. What is the probability of the following events?
   Event A: Rolling a 4.
   Event B: Rolling an even number.
   Event C: Rolling a number greater than 3.

2. You are tossing a fair coin.
   Event A: Getting a Head.
   Event B: Getting a Tail.
   Event C: Getting at least one Head in two tosses.

Normal Distribution

- A bell-shaped curve that shows how data is spread out.
- Symmetrical around the middle (mean).
- Most values are close to the mean, with fewer values toward the edges.
- For example, Heights of people are usually normally distributed—most people are around the average height, with fewer very tall or very short people.
- Mean, Median, and Mode are Equal: All are in the center of the curve.

Z - Score

The z-score (also known as the standard score or z-value) is a statistical measurement that describes a value's relationship to the mean of a group of values. It is expressed in terms of standard deviations from the mean.

$$z = \frac{X - \mu}{\sigma}$$

X = the value (or score) you're analyzing.

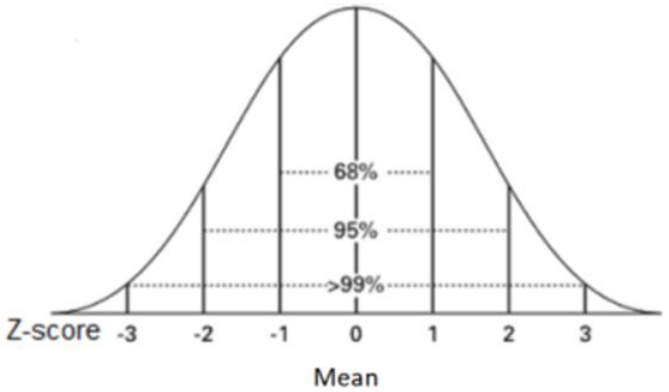
μ = the **mean** (average) of the dataset.

σ = the **standard deviation** of the dataset.

Normal Distribution with Z-Score

- 68-95-99.7 Rule:
  - 68% of data falls within 1 standard deviation (SD) of the mean.
  - 95% falls within 2 SDs.
  - 99.7% falls within 3 SDs.

- Example: If the average exam score is 70:
  - 68% of students scored between 60 and 80.
  - 95% scored between 50 and 90.
  - 99.7% scored between 40 and 100.



Problem Statement

1. A high school tracks the scores of its students in a math exam. The scores are normally distributed with a mean of 70 and a standard deviation of 8.
   The scores of 10 randomly selected students are as follows:
   [85, 62, 75, 91, 68, 80, 67, 72, 95, 58]
   Task 1: Calculate the z-score for each student's exam score.
   Task 2: Identify how many students scored above the average (i.e., have a positive z-score).
   Task 3: Which student has the score farthest from the mean? (This could be either above or below the mean.)

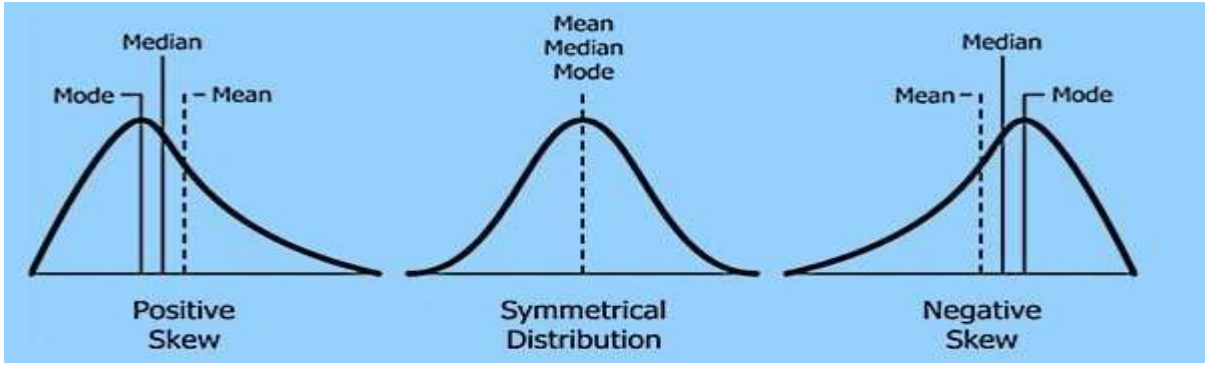
2. A high school tracks the scores of its students in a math exam. The scores are normally distributed with a mean of 70 and a standard deviation of 8. If Sam has a Z-score of 1.5, calculate the Sam's exam score.

Skewness

$$\text{Skewness} = \frac{3(\mu - \text{Median})}{\sigma}$$

- A measure of the asymmetry of the probability distribution of a real-valued random variable.

- **Types:**
  - **Positive Skew (Right Skew):** The tail on the right side is longer or fatter than the left side. Indicates that the mean is greater than the median.
  - **Negative Skew (Left Skew):** The tail on the left side is longer or fatter than the right side. Indicates that the mean is less than the median.
  - **Zero Skew:** The distribution is symmetrical, with the mean, median, and mode being equal.

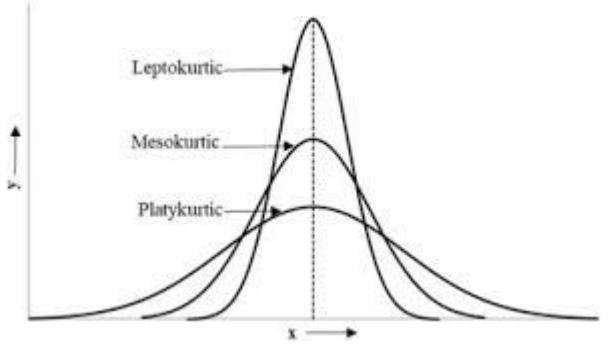


Kurtosis

$$K = \frac{N(N+1)}{(N-1)(N-2)(N-3)} \sum \left(\frac{X_i - \bar{X}}{S} \right)^4 - \frac{3(N-1)^2}{(N-2)(N-3)}$$

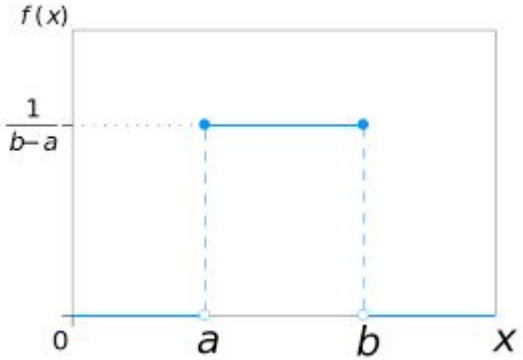
- Kurtosis tells us about the shape of data in the tails of a distribution (the ends of the curve).
- It helps show if data points are concentrated around the mean or spread out in the tails.
- Why it's Important: Kurtosis helps us see how likely we are to have extreme values (outliers).

- **Types:**
  - **Mesokurtic (Normal Shape):** Tails and peak similar to a normal bell curve.
    - Example: Heights of people, where most values are near the average.
    - Kurtosis value: Exactly 3 (or 0 excess kurtosis).
  - **Leptokurtic (Tall & Thin):** High peak, heavy tails—more extreme values.
    - Example: Financial returns, where big gains or losses are common.
    - Kurtosis value: Greater than 3 (often represented as positive excess
  - **Platykurtic (Flat & Wide):** Flatter peak, light tails—fewer extreme values.
    - Example: Dice rolls, with numbers spread evenly.
    - Kurtosis value: Less than 3 (negative excess kurtosis).



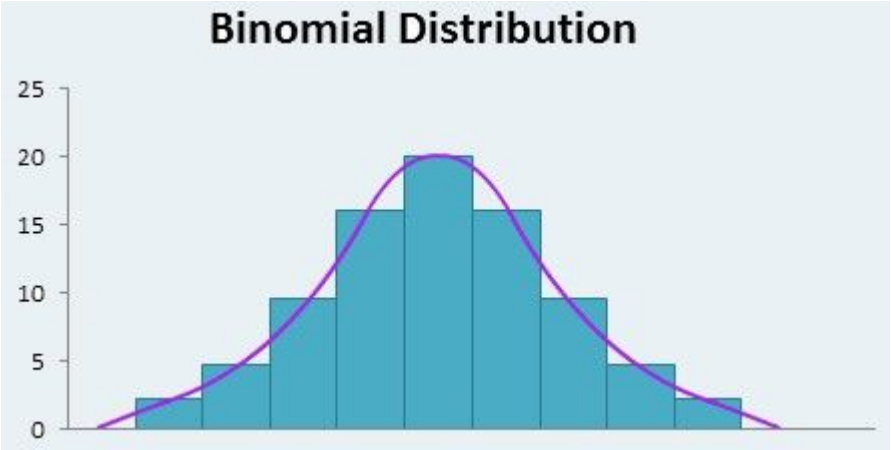
Uniform Distribution

- Uniform distribution is a type of probability distribution where all outcomes are equally likely. In other words, the probability of each outcome in the range is the same.
- Used in scenarios where each outcome is equally likely, such as rolling a fair die or drawing a card from a well-shuffled deck.



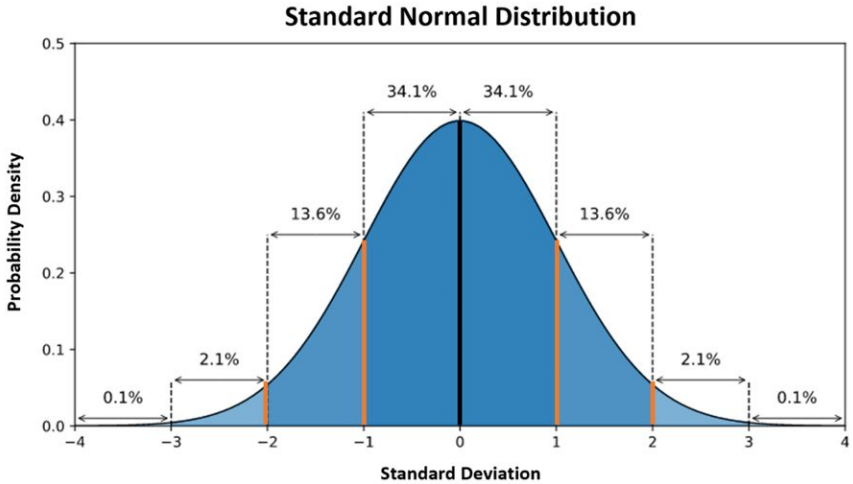
Binomial Distribution

- The binomial distribution models the number of successes in a fixed number of independent trials of a binary (yes/no) experiment. It is characterized by two parameters: the number of trials n and the probability of success p.
- Commonly used in quality control, survey analysis, and any situation where you want to determine the probability of a certain number of successes in a given number of trials (e.g., flipping a coin, testing defective items).



Standard Normal Distribution

- The standard normal distribution is a special case of the normal distribution with a mean of 0 and a standard deviation of 1. It allows for the comparison of different normal distributions by converting their values to a common scale.



Uniform vs Binomial vs Normal Distribution

Aspect	Uniform	Binomial	Normal
Data Type	Discrete or Continuous	Discrete	Continuous
Shape	Flat	Bell-like (may be skewed)	Symmetrical Bell (centered at mean)
Common Uses	Equal chance events	Successes in trials	Naturally occurring data
Examples	Rolling a die, random numbers	Coin flips, defect counts	Heights, test scores, errors
Key Parameters	Min/Max values	Number of trials (n), probability (p)	Mean (μ), standard deviation (σ)

Inferential Statistics

- **Population vs. Sample:**
  - **Population:** The entire set of all possible data points of interest.
  - **Sample:** A subset of the population, used because it's often impractical to gather data from the entire population.
- **Purpose in ML:** Samples allow us to generalize findings from a manageable subset to the larger population, but sampling must be done carefully to avoid bias.
- **Sampling Methods:** Random sampling, stratified sampling, and systematic sampling are common techniques used to ensure representative data.

Sampling Methods

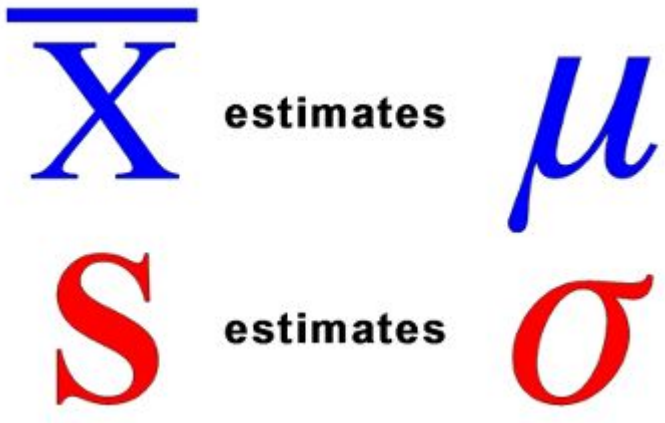
- **Simple Random Sampling:** Every individual in the population has an equal chance of being chosen.
- **Systematic Sampling:** Select every k-th individual from a list after choosing a random starting point.
- **Stratified Sampling:** Divide the population into subgroups (strata) based on a characteristic, then randomly sample from each subgroup.
- **Cluster Sampling:** Divide the population into clusters (groups), then randomly select whole clusters instead of individuals.

Sampling Methods



Estimating Population Parameters

1. **Population Parameters:** These are fixed, often unknown values that describe the entire population. Examples include the population mean (μ), and population standard deviation (σ).
2. **Sample Statistics:** These are calculated from sample data and serve as estimates of the population parameters. Examples include sample mean ($\bar{x}$), and sample standard deviation (s).



Estimating Population Parameters

- Estimating a population parameter involves using data from a sample to make inferences about an unknown characteristic of an entire population, such as the mean, proportion, or variance.
  1. **Point Estimate:** This is a single value that's our best guess for the population parameter.
     - If we measure the heights of 10 students and find the average height is 5.5 feet, then 5.5 feet is our point estimate of the average height for all students.
  2. **Confidence Interval:** Instead of just giving one estimate, we provide a range (called an interval) where we think the true population parameter is likely to be.
     - Suppose our confidence interval is 5.2 to 5.8 feet. This means we are fairly confident the true average height of all students is between 5.2 and 5.8 feet.
     - The most common confidence level is 95%, meaning if we repeated our study multiple times, about 95% of the intervals we calculate would contain the true population parameter.
     - Confidence intervals give us an idea of how accurate our estimate is, instead of just one single number. A narrower interval means a more precise estimate.

Estimating Population Parameters

Confidence Intervals

- A confidence interval (CI) gives a range of values for the parameter, along with a confidence level (like 90%, 95%, or 99%), indicating how sure we are that the interval contains the true parameter.
- For a population mean (when the population standard deviation is known), the confidence interval is calculated as:

$$\text{CI} = \bar{x} \pm Z \left(\frac{\sigma}{\sqrt{n}} \right)$$

$\bar{x}$: sample mean

Z: Z-score corresponding to the desired confidence level (e.g., 1.96 for 95%)

σ: population standard deviation

n: sample size

Estimating Population Parameters

Margin of Error

- The margin of error (ME) represents the maximum expected difference between the true population parameter and the sample statistic.
- It is often expressed in confidence intervals and depends on the sample size, variability, and confidence level.

$$\mathrm{ME} = Z\left(\frac{\sigma}{\sqrt{n}}\right)$$

Hypothesis Testing

- Hypothesis testing is a foundational concept in statistics, helping researchers and analysts make data-driven decisions.

- It is about assessing evidence: Do the sample data provide enough evidence to support a claim about the population?

- Examples of Real-World Applications:
    - Medicine: Testing if a new drug is more effective than an existing one.
    - Business: Determining if a new marketing strategy increases sales.
    - Education: Evaluating if a new teaching method improves student performance.
    - Manufacturing: Ensuring a product meets quality standards.

Steps of Hypothesis Testing

1. State the Hypotheses
2. Set the Significance Level (α)
3. Choose the Appropriate Test
4. Calculate the Test Statistic
5. Find the P-value
6. Make a Decision
7. Draw a Conclusion

State the Hypotheses

- Null Hypothesis (H_0):
  Defined as the hypothesis that there is no effect or no difference.

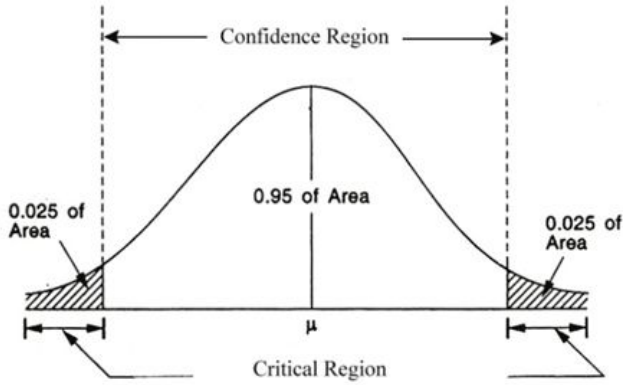
- Alternative Hypothesis (H_1):
  Represents a new effect, difference, or relationship that you want to test.

- **Example:**
  H_0: The average test score is 75.
  H_1: The average test score is not 75.

Set the Significance Level (α)

- The level of significance is the demarcation between the confidence and critical regions in a normal distribution curve.
- The confidence region is that area under the curve in which the null hypothesis is accepted.
- It is also termed as acceptance region.
- The region in which the null hypothesis is rejected is called the critical region or rejection region.
- In a normal distribution curve, 95% of area under the curve is the confidence region and rest is the critical region.

- α=0.05 (5%): This means there is a 5% chance of rejecting the null hypothesis when it is actually true.
- α=0.01 (1%): This means there is a 1% chance of incorrectly rejecting the null hypothesis.



Choose the Appropriate Test

Select the statistical test based on the data type and research question.

1. **t-test:**
   - Used to compare sample means to determine if there's evidence that the population mean differs from a known or hypothesized value, or to compare the means of two groups.
   - **When to Use:**
     - When the sample size is small (typically $n<30$).
     - When the population standard deviation is unknown.
     - When the data is approximately normally distributed.

2. **z-test:**
   - Also used to compare means, particularly when the population mean is known or when comparing proportions.
   - **When to Use:**
     - When the sample size is large (typically $n>30$).
     - When the population standard deviation is known.
     - When the data is approximately normally distributed.

Calculate the Test Statistic

1. **t-test**

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$\bar{x}$ = Sample mean

μ = Hypothesized population mean

s = Sample standard deviation

n = Sample size

2. **z-test**

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$\bar{x}$ = Sample mean

μ = Known population mean

σ = Population standard deviation

n = Sample size

Find the P-value

- The p-value tells you the probability of obtaining the observed results, assuming the null hypothesis is true.
- If the p-value is less than or equal to the significance level (α), you reject the null hypothesis.

1. **t-test:**
    - Determine degrees of freedom (df): $\mathrm{df} = n - 1$
    - Look up the p-value in a t-distribution table or use statistical software.
    - https://www.sjsu.edu/faculty/gerstman/StatPrimer/t-table.pdf

2. **z-test:**
    - Look up the area corresponding to the z-statistic in a standard normal distribution table or use statistical software.
    - https://www.sjsu.edu/faculty/gerstman/StatPrimer/negative-z.pdf
    - https://www.sjsu.edu/faculty/gerstman/StatPrimer/positive-z.pdf

Make a Decision & Draw Conclusions

- Reject H_0: If the p-value $\leq \alpha$, there is enough evidence to reject the null hypothesis.
- Fail to Reject H_0: If the p-value $> \alpha$, you do not have enough evidence to reject the null hypothesis.

- Based on the decision, interpret the results in the context of the research problem. For example:
  - If you reject H_0, you might say "There is sufficient evidence to suggest the average test score is different from 75."
  - If you fail to reject H_0, you might say "There is insufficient evidence to suggest the average test score is different from 75."

Problem Statement

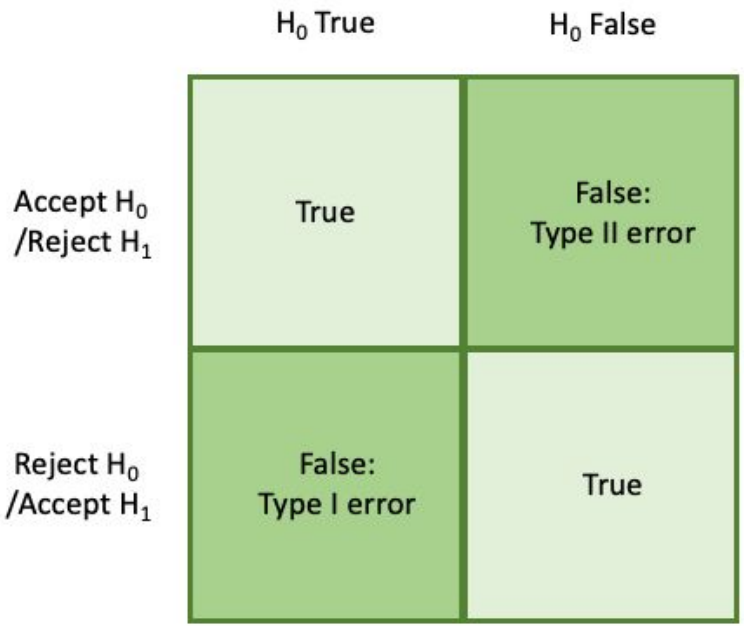
1. A university claims that the average time students spend studying per week is 15 hours. To test this claim, a professor takes a random sample of 25 students and finds their average study time is 13.5 hours, with a standard deviation of 4 hours.

2. A manufacturer claims that the average lifetime of their new LED light bulbs is 1,000 hours. A consumer protection agency tests a random sample of 50 light bulbs and finds that the average lifetime is 980 hours, with a known population standard deviation of 30 hours.

Type I and Type II Error

- **Type I Error (False Positive):**
  - Occurs when the null hypothesis (H_0) is incorrectly rejected when it is actually true.

- **Type II Error (False Negative):**
  - Occurs when the null hypothesis (H_0) is incorrectly not rejected when it is actually false.



Central Limit Theorem

For a sufficiently large sample size, the distribution of the sample mean (or sum) of a random variable will approach a normal distribution (bell curve), regardless of the shape of the population distribution, as long as the samples are independent and identically distributed with finite variance.

1. Data from exponential distribution
2. 1000 samples of size 5
3. 1000 samples of size 10
4. 1000 samples of size 30

